

GHD: minimal note, §1–§2.3

De Nardis–Bernard–Doyon, *Diffusion in generalized hydrodynamics and quasiparticle scattering*.

Scope: introduction, hydrodynamic closure, sum rules, correlation evolution; Appendix B only as needed. Group-discussion synthesis; source-checked and tutor-reviewed.

1. The problem and physical picture [1, §1]

Predict late-time density/current profiles from initial spatial profiles, without evolving the full many-body wavefunction. Conserved densities are slow variables: they redistribute by transport rather than disappear locally. Integrability requires the full set of relevant local/quasilocal charges, conveniently encoded by quasiparticle densities.

$$\partial_t \rho_P + \partial_x (v^{\text{eff}}[\rho_P] \rho_P) = 0, \quad \bar{q}_i = \int d\theta h_i(\theta) \rho_P(\theta).$$

Here one species is shown; θ labels quasiparticles and h_i is their charge. Euler motion has $x \sim t$; diffusion broadens a moving mode on $\Delta x \sim \sqrt{t}$. Interactions already enter Euler GHD through dressing and v^{eff} : Euler does *not* mean noninteracting. The paper links Euler transport to one-particle–hole excitations and diffusion to two-particle–hole scattering, under hydrodynamic and form-factor assumptions.[1]

2. Exact conservation versus closure [1, §2.1]

Use q_i, j_i for microscopic operators and bars for means:

$$Q_i = \int dx q_i, \quad \partial_t q_i + \partial_x j_i = 0, \quad \partial_t \bar{q}_i + \partial_x \bar{j}_i = 0.$$

Q_i is an operator; $\int dx \bar{q}_i = \langle Q_i \rangle$. Local densities need not be time-independent.

After local relaxation, a cell is represented at Euler order by the constrained maximum-entropy state

$$\rho_\beta = Z^{-1} e^{-\sum_i \beta_i Q_i}, \quad \langle q_i \rangle_{\beta(x,t)} = \bar{q}_i(x,t).$$

This follows by extremizing $-\text{Tr} \rho \log \rho$ with fixed charge means and normalization. A *time slice* means fixed t , not time averaging. Closure determines local *expectation values*, not operator identities or an arbitrary microscopic state.

Locality and slow variation motivate (do not prove)

$$\begin{aligned} \bar{j}_i &= \mathcal{J}_i[\bar{q}(\cdot, t)](x) \\ &= F_i(\bar{q}) - \frac{1}{2} \mathfrak{D}_i^j(\bar{q}) \partial_x \bar{q}_j + O(\partial_x^2), \\ \partial_t \bar{q}_i + \partial_x F_i &= \frac{1}{2} \partial_x (\mathfrak{D}_i^j \partial_x \bar{q}_j). \end{aligned}$$

Repeated charge indices are summed. $O(\partial_x^2)$ counts total gradient order, including gradient products. F_i is the homogeneous equation of state; $\mathfrak{D}$ requires transport information. Taylor expansion motivates gradient structure but does not calculate its coefficients. The $1/2$ is a convention, not a Taylor coefficient: for a scalar mode, the usual diffusion constant is $\mathfrak{D}/2$.

3. Which state is held fixed? (discussion branch)

Homogeneity means $T(a)\rho T(a)^\dagger = \rho$, not “no free x in ρ ”. One fixed state $\rho[\beta(\cdot)] \propto e^{-\int dy \beta_i(y) q_i(y)}$ can be inhomogeneous. At fixed global state, $\partial_x \langle o(x) \rangle_\rho = \langle \partial_x o(x) \rangle_\rho$. For

a family $\rho_x = \rho_{\beta(x)}$, instead,

$$\partial_x \text{Tr}(\rho_x o(x)) = \text{Tr}(\rho_x \partial_x o) + \text{Tr}((\partial_x \rho_x) o).$$

Do not switch between these meanings inside a derivation. A homogeneous stationary GGE has stationary means, possibly a nonzero uniform current; an arbitrary homogeneous quench may still have microscopic relaxation.

4. Transport from fluctuations [1, §2.2]

In a homogeneous stationary background, define

$$\begin{aligned} S_{ij}(x, t) &= \langle q_i(x, t) q_j(0, 0) \rangle^c, \\ G_{ij}(x, t) &= \langle j_i(x, t) j_j(0, 0) \rangle^c, \\ K(t) &= \int dx G(x, t), \quad M_n(t) = \int dx x^n S(x, t), \\ C &= M_0(t) = \int dx S(x, 0). \end{aligned}$$

Connected means $\langle AB \rangle^c = \langle AB \rangle - \langle A \rangle \langle B \rangle$. Stationarity fixes dependence on time *difference*; it does not freeze S . Conservation fixes C , not S at each x .

Derivation backbone. Apply continuity at both insertions. Translation invariance gives $\partial_{t'} = -\partial_t$ and $\partial_y = -\partial_x$ on functions of $(x - y, t - t')$ (chain rule, *not* integration by parts). Thus

$$\partial_t^2 S = \partial_x^2 G \implies M_2''(t) = 2K(t),$$

where the latter step integrates by parts twice. For $t > 0$,

$$\frac{M_2(t) + M_2(-t)}{2} - M_2(0) = \int_{-t}^t du (t - |u|) K(u).$$

Define distinct objects (paper Eqs. 2.15–2.16):

$$D = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T du K(u)$$

$$\mathfrak{L} = \lim_{T \rightarrow \infty} \int_{-T}^T du [K(u) - D].$$

D : Drude matrix; $\mathfrak{D}$: diffusion matrix; $\mathfrak{L}$: Onsager matrix; C : susceptibility. If $K = D + k$ with integrable remainder,

$$\frac{M_2(t) + M_2(-t)}{2} - M_2(0) = Dt^2 + \mathfrak{L}t + o(t).$$

A decaying kernel can have a nonzero area; decay alone does not ensure convergence. M_2 is an uncentered correlation moment, not generally a probability variance or merely packet width. Divergent $\mathfrak{L}$ invalidates finite ordinary diffusion.

§2.3: the bridge from closure to correlation spreading

5. Positive times: differentiate means [1, App. B]

All matrices below are evaluated in a fixed homogeneous stationary background. Define the flux Jacobian

$$A_i^j = \frac{\partial F_i}{\partial q_j}.$$

Appendix B assumes initial perturbations $\eta_j(y)$ such that

$$\frac{\delta \langle o(x, t) \rangle}{\delta \eta_j(y)} = \langle o(x, t) q_j(y, 0) \rangle^c.$$

We rename the Appendix's source β as η : its *positive* source convention must not be confused with the minus sign in $e^{-\beta_i Q_i}$. Its existence at diffusive order is an assumption. For noncommuting quantum local sources, differentiation of a Gibbs exponential generally yields a Kubo–Mori correlation, not automatically the ordinary ordered product; do not treat (B.3) as an unrestricted exact identity.

The functional chain rule sums responses over the profile at the *same time* t (not over intermediate times):

$$\begin{aligned} \frac{\delta \bar{j}_i(x, t)}{\delta \eta_j(0)} &= \int dy \frac{\delta \bar{j}_i(x, t)}{\delta \bar{q}_k(y, t)} \frac{\delta \bar{q}_k(y, t)}{\delta \eta_j(0)}, \\ \frac{\delta \bar{j}_i(x, t)}{\delta \bar{q}_k(y, t)} &= A_i^k \delta(x - y) - \frac{1}{2} \mathfrak{D}_i^k \partial_x \delta(x - y). \end{aligned}$$

Varying $\mathfrak{D}$ produces $(\delta \mathfrak{D}) \partial_x \bar{q} = 0$ on the background; this does not make $\mathfrak{D}$ state-independent. $A = \partial F / \partial \bar{q}$ itself is retained. Since $\delta \bar{q}_k / \delta \eta_j = S_{kj}$,

$$\langle j_i(x, t) q_j(0) \rangle^c = (A_i^k - \frac{1}{2} \mathfrak{D}_i^k \partial_x) S_{kj}.$$

Use exact continuity, not an operator substitution:

$$\partial_t S = -A \partial_x S + \frac{1}{2} \mathfrak{D} \partial_x^2 S, \quad t > 0.$$

This is a long-wavelength hydrodynamic equation, not exact microscopic dynamics at every x, t .

6. Negative times: move the later current [1, B.5]

Let $J_{ij} = \langle j_i(x, t) q_j(0) \rangle^c$ and $H_{ij} = \langle q_i(x, t) j_j(0) \rangle^c$. Continuity and translations give

$$\partial_x J_{ij} = -\partial_t S_{ij} = \partial_x H_{ij}.$$

Clustering removes their spatially constant difference, so $J_{ij} = H_{ij}$. Then

$$H_{ij}(x, t) = \langle q_i(0, 0) j_j(-x, -t) \rangle^c.$$

For $t < 0$ the current is now at a later time $-t > 0$. Apply the right-insertion version of the response argument; $\partial_{(-x)} = -\partial_x$ gives

$$J_{ij} = A_j^k S_{ik} + \frac{1}{2} \mathfrak{D}_j^k \partial_x S_{ik},$$

$$\partial_t S = -(\partial_x S) A^\top - \frac{1}{2} (\partial_x^2 S) \mathfrak{D}^\top, \quad t < 0.$$

Translation preserves operator order: it is *not* permission to commute two quantum operators. The generator acts on the second index, hence right multiplication by transposes. With $\tau = -t > 0$, diffusion again has a positive coefficient in $\partial_\tau S(x, -\tau)$; this is not inversion of a dissipative initial-value problem. The last displayed formula of

Appendix B must contain S_{ik} , not the printed S_{ij} under the k sum.

7. Spatial moments identify transport [1, §2.3]

Keep the initial first moment:

$$E = M_1(0) = \int dx x S(x, 0), \quad AC = CA^\top.$$

A need not be symmetric. If $C > 0$ is invertible, A is self-adjoint in the C^{-1} metric; equivalently $C^{-1/2} A C^{1/2}$ is symmetric, giving real mode velocities.

Integration by parts yields

$$\begin{aligned} M_0 &= C, & M_1(t) &= ACt + E, \\ M_2'(t) &= 2AM_1(t) + \mathfrak{D}C & (t > 0). \end{aligned}$$

For $t > 0$, the two integrated branches are

$$\begin{aligned} M_2(t) - M_2(0) &= A^2 C t^2 + (2AE + \mathfrak{D}C)t, \\ M_2(-t) - M_2(0) &= A^2 C t^2 + (C \mathfrak{D}^\top - 2EA^\top)t. \end{aligned}$$

These equalities are within the retained hydrodynamic approximation. Averaging and matching §2.2 gives

$$D = A^2 C = AC A^\top,$$

$$\mathfrak{L} = \frac{1}{2} (\mathfrak{D}C + C \mathfrak{D}^\top) + AE - EA^\top.$$

Source discrepancy. Eqs. (2.26–2.27) print only $\frac{1}{2}(AE - EA^\top)$. Direct integration of (2.21), (2.24–2.25), with (2.22)'s definition of E , gives the full coefficient above. This is an apparent factor-of-two typo; the note uses the displayed derivation, not a verified author erratum.

Scope boundary. Do not yet assume $E = 0$ or $\mathfrak{D}C = C \mathfrak{D}^\top$. If both hold, then $\mathfrak{L} = \mathfrak{D}C$ and, on an invertible sector, $\mathfrak{D} = \mathfrak{L}C^{-1}$. Their gauge/PT justification belongs to §2.4, not a general conclusion of §2.3.

8. Validity and three recall checks

Assume local relaxation, a complete slow-variable description, slow gradients, homogeneous stationary background for correlations, sufficient spatial decay for moments/boundary terms, and convergent transport limits. Neither microscopic reversibility nor conservation alone proves diffusive closure.

Q1. Why t^2 versus t ? **A.** Drift acts on $M_1 \propto t$; diffusion acts on $M_0 = C$.

Q2. Why subtract D from K ? **A.** Remove persistent ballistic correlations before integrating the decaying part.

Q3. Why not substitute $j = F(q) - \mathfrak{D} \partial_x q / 2$? **A.** This closes means, not operators; a source response and chain rule are required.

Source and provenance

[1] J. De Nardis, D. Bernard, B. Doyon, *SciPost Phys.* **6**, 049 (2019), arXiv:1812.00767v4; §1–2.3 and Appendix B. Equations checked against the retrieved source. Discussion-specific caveats retained from the learning group; recovery and audit details are in the companion review file.